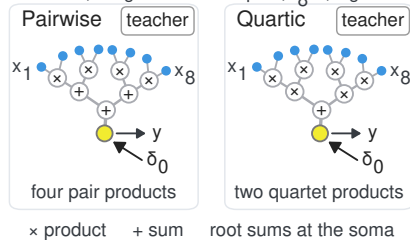
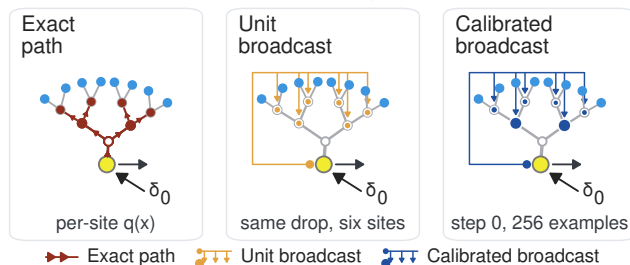


A

Two targets, one shared tree

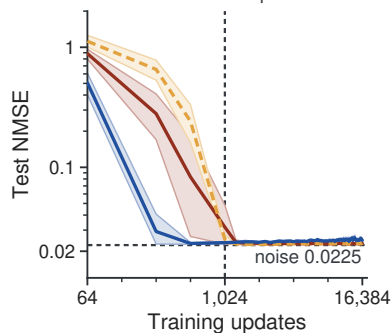
same tree, weights and examples; $\log_2 4$; signs ± 0.5 **B**

Where each rule delivers credit

six nonsomatic sites; radius = mean $|\text{weight}|$ over 20 seeds, 0.133–0.386**C**

Pairwise: broadcast fits

frozen Adam rates

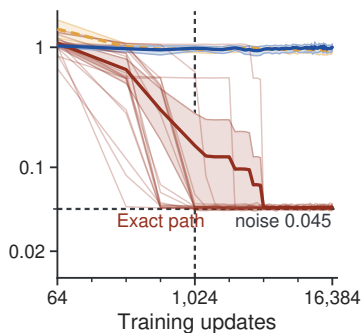
 $n = 20$ paired seeds**D**

Quartic: only exact fits

3 of 20 exact seeds stall

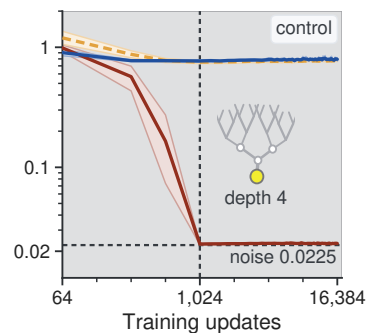
all at floor by 4,096

broadcast and calibrated broadcast

**E**

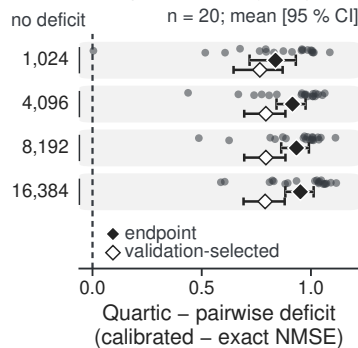
Nested control: depth 4

depth 4, unmatched spectrum

**F**

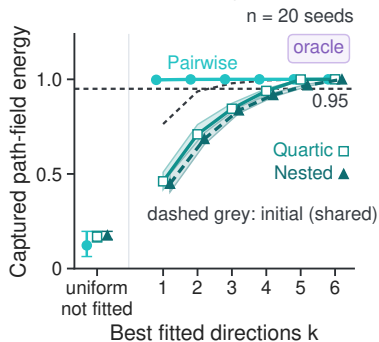
Larger quartic deficit

20/20 positive at every budget

 $n = 20$; mean [95 % CI]**G**

One direction, or five

oracle fit at 1,024 updates

 $n = 20$ seeds**H**

Leaf assignment matters

tree shape, parameters fixed

 $n = 20$; mean [95 % CI]